



UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

Engineering, Built Environment and IT
Department of Computer Science
Imperative Programming
COS 132

Aegrotat for EO1 and EO2

5 June 2023 (14:00 to 15:30)

Name (Initials and Surname): _____

Student number: _____

Degree programme (BSc CS/IKS/Other, BEng, BISMM, Other (specify): _____

Examiners

Internal: Mr Pula Rammoko, Mr Sheanesu Makura, Ms Seani Rananga, Prof Stefan Gruner and Dr Linda Marshall

Instructions

1. Read the question paper carefully and answer all the questions.
2. The assessment opportunity comprises of **7** questions on **9** pages. Refer to the table below for the marks per question.
3. One and a half (**1.5**) hours have been allocated for you to complete this paper.
4. In the event of loadshedding, extra time will be given depending on how long it takes the generators to kick in.
5. You may bring your own notes (both hand written or typed) into the examination room.
6. Write all your answers in the spaces provided on the question paper. All answers must be written indelible pen.
7. You may detach the last page from the paper and use it for rough work.

Integrity statement:

The University of Pretoria commits itself to produce academic work of integrity. I affirm that I am aware of and have read the Rules and Policies of the University, more specifically the Disciplinary Procedure and the Tests and Examinations Rules, which prohibit any unethical, dishonest or improper conduct during tests, assignments, examinations and/or any other forms of assessment. I am aware that no student or any other person may assist or attempt to assist another student, or obtain help, or attempt to obtain help from another student or any other person during tests, assessments, assignments, examinations and/or any other forms of assessment.

Question:	1	2	3	4	5	6	7	Total
Marks:	12	12	9	3	8	6	5	55
Score:								

Full marks is: 50

Questions

1. For each of the questions that follow, choose the most appropriate option by circling the corresponding letter.

(a) Given the following C++ program:

(2)

```
#include <iostream>
using namespace std;

void foo() {
    static int x = 0;
    int y = 0;
    cout << "x = " << ++x << ", y = " << ++y << ";
}

int main() {
    for (int i = 0; i < 3; i++) {
        foo();
        cout << endl;
    }
}
```

What is the output of the program?

- A. x = 0, y = 0; x = 0, y = 0; x = 0, y = 0;
- B. x = 0, y = 0; x = 1, y = 1; x = 2, y = 2;
- C. x = 1, y = 0; x = 1, y = 1; x = 1, y = 2;
- D. x = 1, y = 1; x = 1, y = 2; x = 1, y = 3;
- E. x = 1, y = 1; x = 2, y = 1; x = 3, y = 1;
- F. x = 1, y = 1; x = 1, y = 1; x = 1, y = 1;

(b) Which one of the following is a correct function header for a function that computes and returns the square root of a float value?

(2)

- A. void squareRoot(float value)
- B. float squareRoot(int value)
- C. int squareRoot(float value)
- D. float squareRoot(float value)

(c) Which of the following statements about arrays are true?

(2)

- (1) An array may contain elements of different data types.
 - (2) A constant array must have an initialisation list.
 - (3) When an array is passed as an argument to a function, the function has access to the original array.
- A. (1), (2) and (3) are true.
 - B. (2) and (3) are true.
 - C. (1) and (2) are true.
 - D. Only (3) is true.

(d) Given the following array declaration:

(2)

```
const int SIZE = 5;
int numbers[SIZE] = { 10, 15, 20, 25, 30 };
```

Which of the following statements will display all the values in the array?

- A.

```
for (int index = 1; index < SIZE; index++)
    cout << numbers[index] << '\t';
```
- B.

```
for (int index = 1; index <= SIZE; index++)
    cout << numbers[index] << '\t';
```
- C.

```
for (int index = 0; index < SIZE; index++)
    cout << numbers[index] << '\t';
```

```
D. for (int index = 0; index <= SIZE; index++)  
    cout << numbers[index] << '\t';
```

(e) Which of the following is a correct declaration for a 2-dimensional array with 5 rows and 10 columns? (2)

- A. `int table[5][10];`
- B. `int table [10][5];`
- C. `int table [5], [10];`
- D. `int table [10, 5];`

(f) Given the following declarations: (2)

```
float num1 = 10.5 , num2 = 5.2 , temp;
```

Which of the following statements will correctly swap the values of num1 and num2;

- A. `num1 = num2;`
`num2 = num1;`
- B. `num2 = num1;`
`num1 = num2;`
- C. `temp = num1;`
`num1 = num2;`
`num2 = temp;`
- D. `temp = num2;`
`num2 = temp;`
`num1 = num2;`

2. Given the following C++ program. Answer the questions that follow.

```
1 #include <iostream>
2 using namespace std;
3
4 int main() {
5     int num = 0;
6
7     cout << "Enter a number from 1 to 3: ";
8     cin >> num;
9     switch (num) {
10         case 1:
11             cout << "Case 1" << endl; break;
12         case 2:
13             cout << "Case 2" << endl;
14         case 3:
15             cout << "Case 3" << endl; break
16     }
17     return 0;
18 }
```

(a) Identify the syntax error in the given C++ program. The program is intended to use a **switch**-statement to print different messages based on the value(s) of the variable **num**. However, there is an error in the code. Identify the line number of the error and explain what needs to be fixed to resolve it. (2)

(b) What could the output of the program be if 2 is entered when prompted for a number? You may assume the syntax error in (a) has been fixed. (2)

(c) Update the code to make provision for numbers entered that are not 1, 2 or 3. The program should print the words "Incorrect input" to the screen. (2)

Write the code next to the given program listing and indicate after which statement in the given code your code needs to be included.

(d) Rewrite the given **switch**-statement as an **if**-statement. (4)

(e) Under what conditions can an **if**-statement be converted to a **switch**-statement? (2)

3. Consider the program given below and answer the questions that follow.

```
1 #include <iostream>
2
3 using namespace std;
4 int main() {
```

```

5   int aa = 2;
6   while (aa <= 4) {
7       int bb = 2;
8       while (bb <= 3) {
9           cout << aa << " " << bb << endl;
10          bb++;
11      }
12      aa++;
13  }
14 }

```

(a) When the above program is run, what will be the output?

(6)

(b) Rewrite the inner **while**-statement as a **for**-statement. Specify the lines of code in the given program that need to be changed and provide the replacement code. (3)

4. Complete the following program which must use a **while** loop, to print all even numbers between 10 and 30 (inclusive). (Only include the **while** loop) (3)

```
#include <iostream>
using namespace std;

int main() {
    int number = 10;

    cout << "Even numbers between 10 and 30 are: ";

    //The code for the while loop should be written here

    return 0;
}
```

5. Study the following C++ code. The code manipulates an array of size **s**. The first loop assigns the value of **t** to each element of the array. The second loop sums the elements of the array using variable **z** and then calculates the average of the elements in the array. (8)

The code contains two errors Identify the errors and provide the line on which each error occurs, an explanation of why it is an error, and how the error should be fixed. Answer the question by completing the table below.

```
1 #include <iostream>
2 using namespace std;
3
4 int main() {
5     const int s = 5;
6     int arr[s];
7     for (t = 0; t < s; t++) {
8         arr[t] = t;
9     }
10    int z = 0;
11    for (int r = 0; r <= s; r++) {
12        z = arr[r] + z;
13    }
14    cout << z/5;
15    return 0;
16 }
```

	Error 1	Error 2
Error type		
Line number		
Explanation		
Fix		

6. Study the following C++ code and answer the questions that follow:

```
1 #include <iostream>
2 using namespace std;
3
4 int main () {
5     int year1 = 2023;
6     int year2 = 2024;
7     cout << year1 << endl;
8     cout << year2 << endl;
9     cos132(year1, year2);
10    cout << year1 << endl;
11    cout << year2 << endl;
12    return 0;
13 }
14
15 void cos132(int &bb, int &cc) {
16     int aa;
17     aa = bb;
18     bb = cc;
19     cc = aa;
20 }
```

(a) As given, the program will not compile. Identify the problem. (1)

(b) What is the name of the parameter type, defined with a &, and used in line 15? (1)

(c) What is the purpose of the parameter you have identified in (b)? (2)

(d) When the above program is run, what will be the output? You may assume the error has been fixed. (2)

7. Consider the code segment given below that defines a two dimensional array and calls the function `print_2Darray`. (5)

```
const int ROWS = 3;
const int COLS = 4;

int arr[ROWS][COLS] = {
    {1, 2, 3, 4},
    {5, 6, 7, 8},
    {9, 10, 11, 12}
};

print_2Darray(arr, ROWS, COLS);
```

Write function `print_2Darray`. For the array given in the code segment, the tab-delimited values per row written to the screen would be:

1	2	3	4
5	6	7	8
9	10	11	12